

Energy measurement with ICs based on autocalibration and non-autocalibration

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Abstract—The public lighting system (PLS) is an essential mechanism in the advancement of city technology in the current context, serving not only a visual purpose but also aiding in the safety and guidance of the population. Therefore, the current PLS faces several challenges today. In addition to modernizing old high-intensity discharge (HID) lamps to light-emitting diodes (LEDs), there are also smart measurement and innovations related to the maintenance of luminaires that are damaged or not functioning. Thus, a smart and accurate measurement system is necessary to follow this path. Therefore, this work aimed to implement a measurement system with sensors with and without self-calibration in MATLAB to understand the impacts of this functionality on measurement accuracy. The data used were based on the ADE9153A IC.

Index Terms—Energy meter, Matlab, measurement

I. INTRODUCTION

Electricity consumption increases by about 3%, with 15–19% of all this energy being used for public lighting systems. Such energy consumption accounts for about 5–6% of all greenhouse gas emissions to the atmosphere [1]. Expenditures related to the current public lighting system based on high-intensity discharge (HID) luminaires are mostly due to maintenance and operation rather than implementation, as there are many losses intrinsic to such equipment. Therefore, lighting based on light-emitting diode (LED) lamps can lead to a significant reduction in energy costs, around 30% of expenses, and with the use of auxiliary equipment and intelligent control, it can achieve impressive savings of 80% [2].

In addition to being used as a source of illumination, this new system can explore various possibilities such as dynamic dimming for energy savings, fault detection mode for the maintenance of defective luminaires, communication with public safety sensors to reduce crime, data collection regarding pedestrian and vehicle flow for the sale of traffic information, among others. However, this requires a metrology and telemangement system for these luminaires, as there must be an entire electronic setup for executing these commands. Thus, this telemangement equipment must meet certain standards related to its functionality; in Brazil, INMETRO Ordinance No. 221 of 2022 is in effect for such systems, where it describes technical operational limits and capabilities [3].

This work aims to use MATLAB software to implement a code for analyzing energy meter integrated circuits to measure

errors related to temperature drift, variation in measurement resistors, offset and gain errors, network instability, and the real sampling frequency of the component. The study will focus on the ADE9153A integrated circuit from Analog Devices, which features autocalibration. The project will be divided into two parts: one without the use of autocalibration or software-based error compensation techniques, and another applying these approaches to enhance the performance and accuracy of the meters.

II. CURRENT SCENARIO OF THE PUBLIC LIGHTING SYSTEM

The charge for public lighting in Brazil is regulated by the Federal Constitution of 1988, which establishes the competence of municipalities to organize and provide, directly or under a concession or permission regime, public services of local interest, including public lighting [4]. This service is essential to ensure safety, mobility, and quality of life in urban areas. COSIP is regulated by municipal legislation, meaning each municipality has the autonomy to define the rates and the method of calculating the contribution. Generally, this charge is proportional to electricity consumption, and this amount is exclusively allocated to the costs of operating, maintaining, and expanding the public lighting system [5]. Municipalities often establish public-private partnerships to modernize and expand public lighting, with the aim of serving more citizens and making the system more efficient, such as with LED-based lighting technology.

Therefore, the current energy billing and measurement system relies much more on maximum estimates than actual values, besides the fact that detecting faulty luminaires is not possible. Hence, a measurement system capable of conducting such measurements would be useful to enhance the efficiency of the public lighting system, both in repairs and in the use of dimming to reduce electricity consumption, among other benefits, such as greater transparency regarding the use of public funds, remote reading of operational values, the possibility of variable tariffs according to monthly usage, and verification of the performance of the installed system.

However, this system can exhibit levels of inaccuracy due to external interferences such as electromagnetic interference, temperature, and humidity, as well as internal system factors

like offset voltages, gain errors, component variations, power supply distortions, and noise. These factors contribute to inaccuracies, causing the system to deviate from the true measurements. While such discrepancies are inevitable, the key concern is the reliability of the measured value in relation to the actual value, specifically how close the measured value is to the true value to be considered acceptable.

In this paper, section III the measurements involved in a measurement system based on an integrated energy meter (CI energy meter) will be discussed. Section IV will discuss the sources of errors affecting the system and propose methods for mitigating these effects. Section V will present discussions on the developed code and its results. Finally, Section VI will offer the conclusions of the study based on all the topics discussed.

III. MEASUREMENTS

A. Current

The accurate measurement of current and voltage values in a metering system will determine its overall precision, as all other metrics will be derived from these values. Therefore, the correct selection of precise components with excellent sensitivity to the range of values to be used is necessary for a viable energy metering project. Measuring current is more challenging than measuring voltage due to the wide dynamic range of amplitude and the processing of a wide range of frequencies present in the current signal [6].

For current measurement, we can use components such as shunt resistors, current transformers (CTs), or Hall effect sensors. CTs are used in high power applications and when galvanic isolation is required. Additionally, the measurement losses are almost negligible and there is no induced voltage at zero current. However, the costs are higher compared to resistive sensors, and there are errors associated with the inaccuracy of the actual transformation ratio and phase shift due to the limited magnetic permeability of the core and the non-zero value of the load resistance [7]. Hall-effect-based sensors tend to be excellent for current measurement due to their high linearity, low loss levels, and near absence of temperature drift and current shift. However, these components are sensitive to external magnetic fields and nearby currents. Additionally, their accuracy is associated with the location of the conductor within the sensor, making the precision and reliability highly variable for industrial use [8].

Shunt resistors emerge as protagonists in a variety of projects due to a combination of factors. Their advantages include low cost, high linearity, impeccable phase integrity at low frequencies, compact dimensions, and remarkable temperature coefficient stability, positioning them as the component of choice for a diverse range of manufacturers, even in the face of the inherent voltage drops in the device [6].

B. Voltage

Voltage measurement can be performed in two main ways: analog or digital. Analog sensors consist of an electronic amplifier and an electromechanical meter to measure the

amplifier's output. This configuration allows the sensor to have a wide bandwidth and high input impedance. However, the weakness of this type of sensor lies in the mechanical parts present, which make the circuit susceptible to external changes and deterioration over time, affecting its reliability and accuracy.

Digital voltage measurement involves quantizing and discretizing the signal to enable digital measurement. This process generally uses a sensor to establish an acceptable voltage level and an analog-to-digital converter (ADC) responsible for transforming the input voltage into a sequence of sampled values. To improve the accuracy of this type of sensor, it is crucial that it has a sampling rate, conversion speed, dynamic range, and resolution appropriate for the system to be measured, as these parameters directly affect the accuracy of the measurement [9].

For voltage sensing, there are various techniques such as using voltage dividers with resistors and capacitors, and transforming the voltage to acceptable levels that can be read and quantized by an IC or microcontroller. Although the use of transformers is effective, it imposes a high cost and the transformation ratio is difficult to maintain constant. Additionally, the presence of DC components in the measured signal can cause problems in the circuit. On the other hand, capacitor-based voltage dividers present challenges such as dependency on temperature, frequency, and component degradation over time. Furthermore, the variation in component values can cause significant distortions in the readings [9]. Voltage dividers using resistors, on the other hand, offer high precision and thermal stability. They also have the advantage of not introducing phase shifts, measuring DC, and are low cost. However, the main disadvantage of this method is the inherent power losses in the measurement system, which occur throughout the entire measurement process.

C. Power

Power is defined as the work done over time. In measuring this type of quantity in AC networks, the integration of the instantaneous value of current and voltage over an interval is used to obtain the *Root Mean Square* (RMS) value of these signals. With the RMS values of current and voltage, it is possible to directly multiply these values to define the power in cases of purely resistive loads. However, in most circuits, there is a phase difference between current and voltage, requiring the multiplication of the phase angle between them. Therefore, Equation 1 demonstrates the relationship for power measurement for any type of load.

$$P = V_{\text{rms}} \times I_{\text{rms}} \times \cos(\theta) \quad (1)$$

Where:

- P is the active power (in watts).
- V_{rms} is the RMS voltage (in volts).
- I_{rms} is the RMS current (in amperes).
- θ is the phase angle between voltage and current.

D. Total Harmonic Distortion (THD)

The Total Harmonic Distortion (THD) is a metric that indicates the fidelity of a system in reproducing an input signal. While a completely sinusoidal wave contains only one frequency component, systems with nonlinearities introduce distortions, resulting in multiple frequency components of the fundamental frequency, known as harmonics. These harmonics alter the waveform, deviating it from its original sinusoidal shape [9].

The THD quantifies this distortion by summing the values of the harmonics in relation to the measurement of the fundamental frequency. Ideally, it would be necessary to measure all harmonics, but as they are infinite, standards establish a limit on the number of relevant harmonics to be considered. To obtain the distribution of waveform components, it is necessary to transform the signal from the time domain to the frequency domain, usually done using the Fast Fourier Transform (FFT). Equation 2 is then applied to define the influence of THD on the system.

$$THD = \frac{\sqrt{\sum_{k=2}^N a_k^2}}{a_1} \times 100\% \quad (2)$$

Where a_1 is the amplitude of the fundamental component and $a_2, a_3, \dots, a_N$ are the amplitudes of the harmonic components.

IV. ERRORS SOURCES

A. Parametric tolerance of components

Electronic components have small parametric differences due to manufacturing uncertainties, resulting in slight variations in values among such elements, causing inaccuracies in circuits. Manufacturers provide maximum variations in datasheets for these components, thus, the choice of components becomes a cost-benefit variable. However, there is the possibility of reducing this error to minimum values, which involves the use of individual equipment calibration, where the IC or microcontroller uses a known reference to analyze the sensor result and calibrate itself to accommodate the parametric variations of the sensor used [10].

A classic calibration method involves employing a shunt resistor in conjunction with a current source as a reference. In this approach, the voltage across the resistor is measured and compared to a known standard value. Subsequently, a multiplier is utilized to adjust the resistor's value to achieve the desired output.

B. Input offset error

The input offset voltage occurs due to the impossibility of creating a completely symmetrical component, resulting in a voltage imbalance between the input terminals. Ideally, to eliminate this value, it would be necessary to reduce it to zero. However, due to design limitations and process variations, this is extremely difficult to achieve. This imbalance can cause large amplitude errors, especially in the presence of small signals. A good example of this is the zero-crossing of the

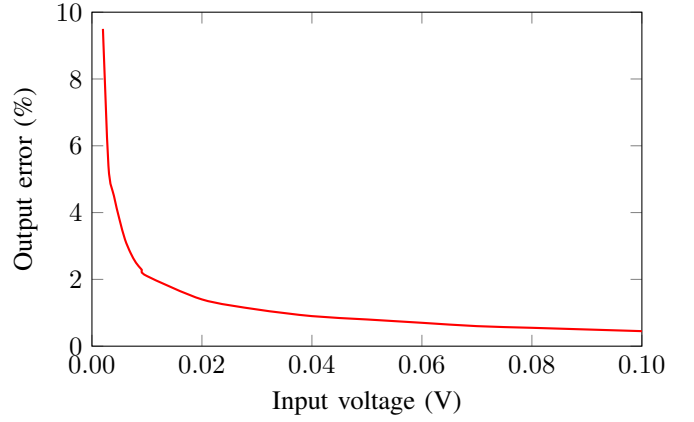


Fig. 1. Example of input offset error compared to input voltage, adapted from [10]

current in a shunt resistor: since the actual value is zero, any input offset error will produce a significant error, as shown in equation 3 of the error [10], [11].

$$Error = \frac{|Measured_Value - True_Value|}{True_Value} \times 100\% \quad (3)$$

Figure 1 also illustrates the predominance of the input offset voltage error at low values read by the sensor.

C. Gain error

The gain error refers to a variation in gain during operation, not being entirely linear across the amplification range. This discrepancy in amplification occurs due to component tolerances, external interference, and electrical imbalances in the system. Being present throughout the signal's extent, the gain error tends to be more significant at larger amplitudes, where its influence is more prominent [9], [10].

Thus, small variations in gain due to improper adjustments can have significant impacts on the measurement circuit, affecting its accuracy and reliability. It's also worth noting that input offset errors will be amplified by the gain, further distorting the signal, especially if it has a very low input voltage range. Figure 2 illustrates the gain variations across the signal, compared to the ideal gain line.

D. Temperature

Temperature has a comprehensive impact on electronic systems, resulting in material degradation, risks of premature failures, and a reduction in component lifespan. Additionally, temperature significantly influences other types of errors, such as component tolerance variation. For instance, temperature changes can cause variations in resistance, capacitance, and inductance values, impairing linear operation and measurement accuracy. These changes are often quantified in parts per million per degree Celsius (ppm/°C), and it's preferable to use critical components with low levels in this measure, as they are less susceptible to temperature variations [8].

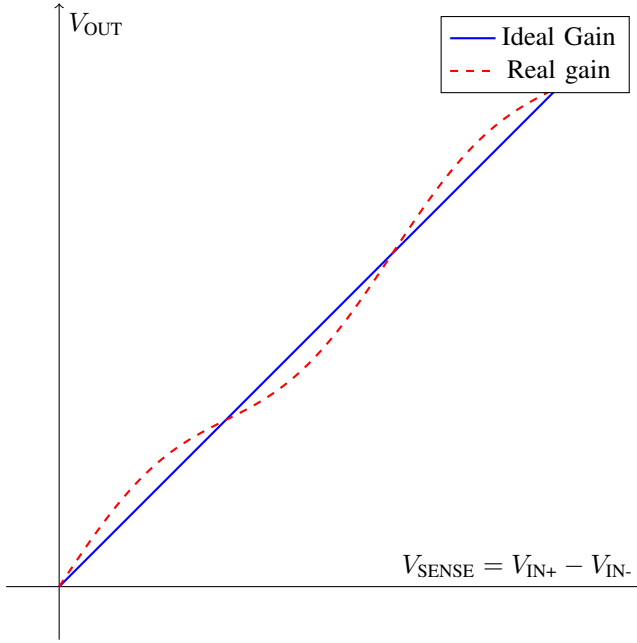


Fig. 2. Ideal gain and real gain, adapted from [10].

Moreover, temperature also contributes to an increase in input offset voltage drift, resulting in greater imprecision stemming from this phenomenon. Concerning signal gain error, it is generally correlated with temperature variations. However, efficient temperature control can significantly reduce this error, enabling better performance than indicated in the component specifications. Thus, proper thermal design of the measurement system is crucial, as its inefficiency compromises the accuracy, reliability, and overall lifespan of the system [10].

V. SIMULATION AND ANALYSIS OF RESULTS

A. System to be analyzed

The system to be analyzed is the distribution network itself, which was simulated as an ideal wave with a voltage value of 220 Vrms and an operating frequency of 60 Hz. This will ensure that the initial analysis conditions are known and errors are isolated. The load will have resistive behavior, consuming an average of 5 A at the nominal voltage of 220 Vrms. The chosen energy meter is the ADE9153A from Analog Devices due to its self-calibration capability. For current measurement, a shunt resistor will be used, and for voltage measurement, a resistive voltage divider. The schematic of the simplified system can be seen in Figure 3.

B. Energy meter ADE9153A

The ADE9153A IC from Analog Devices was chosen for its unique autocalibration feature, which will be used to enhance the measurement circuit's accuracy. Other notable features include low temperature drift, immunity to power supply variations, high sampling rate (up to 8 kSPS), and high signal gain. In addition to these capabilities, the IC includes an integrated temperature sensor for error compensation and the

TABLE I
ADE9153A MEASUREMENT CHANNELS, ADAPTED FROM [12].

Parameter	Typical	Maximum
Current Channel A		
Input offset error (mV)	0.04	0.1
Gain (%)	0.2	1.5
Drift (ppm/°C)	5	30
PSRR (dB)	-115	-
Voltage Channel		
Input offset error (mV)	0.35	0.75
Gain (%)	0.8	3
Drift (ppm/°C)	20	50
PSRR (dB)	-100	-

zero-crossing function detects when the current and voltage signals pass through zero, allowing for the reading of variable frequencies and adjusting via software for better circuit accuracy. The table I presents the key data for the current and voltage measurement channels of the ADE9153A [12].

Other important data for defining the measurement circuit include the dynamic range over which the circuit will perform measurements. In the current measurement channel with a shunt resistor, the input voltage is supported from $-1/\text{gain}$ to $+1/\text{gain}$. Using a gain of 38.4 times, which is the highest possible amplification, the maximum current used will be 5A average. Therefore, the highest voltage amplitude in this channel is 18.41 mV, allowing the use of a 2mΩ shunt resistor while maintaining a wide safety margin.

For the voltage channel, the limits are 353.6mVrms, considering a maximum input voltage of 240Vrms. Applying a resistive divider with a ratio of 1 to 1000 is adequate and still maintains a safety margin for potential network transients.

C. Simulation

To conduct the simulation, a simulated network of 220Vrms and 60Hz over 10 seconds was generated, which will be used for analysis. Additionally, a resistive load was used to maintain an average output current of 5A. These collected values will serve as a reference for comparison with subsequently acquired data. For error analysis, the selected resistors will vary by 1%, and there will be a temperature drift of 50 ppm/°C, spanning temperatures from -40 to 85°C, within the temperature range supported by the IC. Other errors will be randomly varied based on the values specified in the ADE9153A datasheet, as shown in Table I.

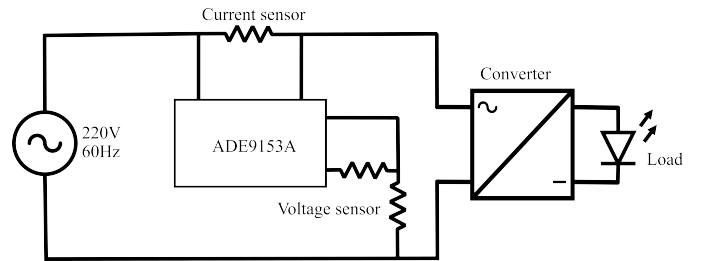


Fig. 3. Circuit schematic.

After acquiring errors without calibration, autocalibration and the use of IC tools such as the temperature sensor will be performed. This implementation aims to minimize errors related to thermal drift, accuracy, offset, and gain. To ensure more consistent data analysis, the simulation will be run a total of 100,000 times and will be represented in the form of a histogram.

For voltage and current signal sensing, resistors are used to provide the same dynamic behavior as the original wave to be measured but with a lower amplitude. The conversion is performed using Ohm's Law, with a ratio of 500:1 for current and 1000:1 for voltage. The errors at this stage are offset, component variation, and temperature drift. However, for voltage measurement, since the relationship between the resistors in the resistive divider is important, if similar and close resistors are used, the effect of temperature drift can be ignored as the relationship remains constant over the entire range.

With the data input at acceptable amplitudes for the IC, an amplification stage is performed to make the signal less susceptible to noise. The current channel amplification in this work is 38.4 times, while the voltage channel amplification is 1 time. The errors that interfere at this stage are the gain errors present in the gain amplifier of the energy meter itself.

Thus, the sampling of the signals is carried out by the AD converter. If the sampling frequency is not at least twice the frequency of the waveform, it is impossible to measure it accurately (Nyquist criterion). Since it is necessary to measure up to the 39th harmonic, the sampling frequency should be 2.34 kHz, with the minimum actual sampling frequency being 4.68 kHz, which is achieved by the ADE9153A, which has a sampling frequency more than three times the required.

With the discretized data, it is transferred to the micro-controller responsible for reading, digitally adjusting the gain for correct readings, calculating other necessary parameters, and finally reconstructing the sampled signals. In Figure 4 the distribution of these errors for current, voltage and power is displayed.

If done this way, this IC would not have the potential to be used for measuring electrical parameters in public lighting. However, with the self-calibration function, it can minimize many of its errors. To do this, initially, a measurement is taken before the amplification stage and without any circulating energy, allowing the offset present in the circuit to be found and subtracted.

Next, a constant known current or voltage signal is applied. This signal will not be amplified as the goal is to measure the real value of the components used. By Ohm's Law, the real values are read and compared with the desired ones, applying a gain to the signal to compensate for imperfections. Finally, the last calibration is performed with the same signal, with all adjustments made, and amplification carried out. This step adjusts the gain of the amplifier, making it as close as possible to the value specified in the datasheet. Disregarding the effects of temperature on offset and gain nonlinearity, the only remaining error in the circuit will be the variation

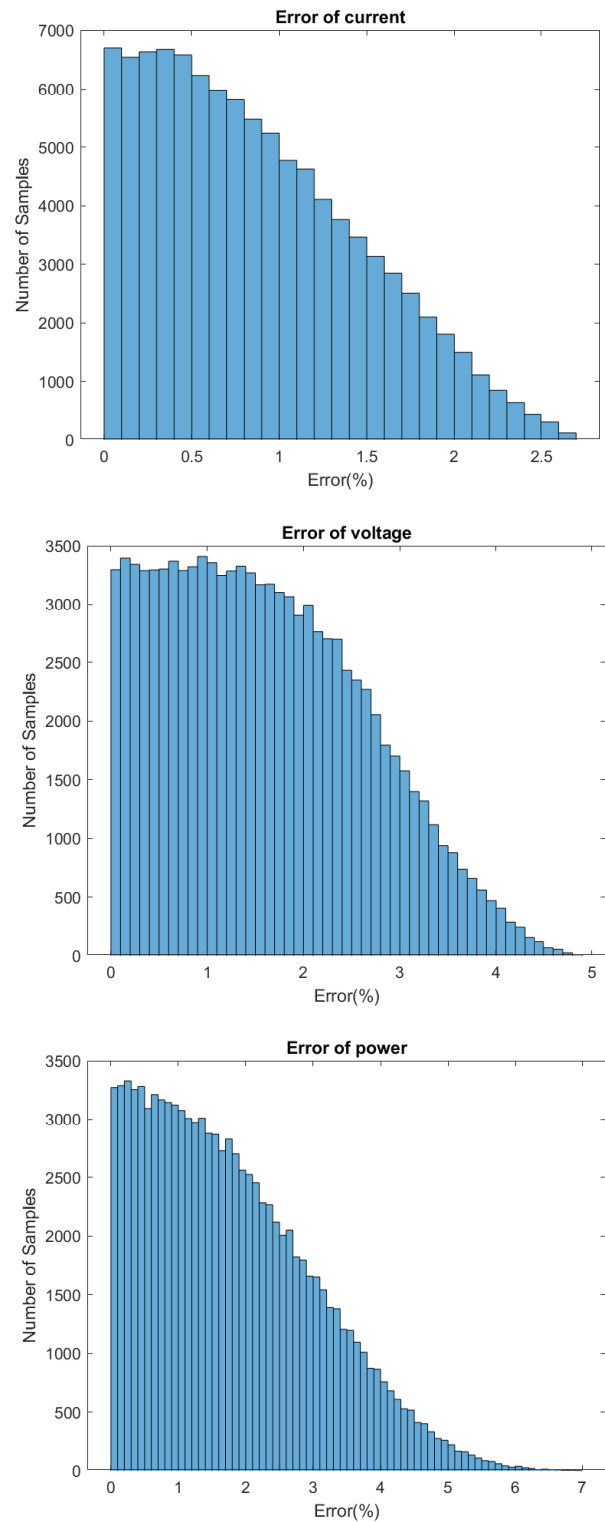


Fig. 4. Errors of current, voltage and power.

of the sensor component due to temperature. However, as mentioned, this effect does not occur in the voltage sensor, so the errors will be very close to zero. In the case of current,

the remaining error will be entirely temperature-dependent and can be minimized by using a resistor with a lower ppm/°C. The results of the errors after calibration are shown in Table II. Since the temperature variation was the same during all tests, the values will always be the same.

In both cases, the THD of the signal was so close to zero that it is practically irrelevant to the waveforms. This occurs due to the linear behavior of the load. If a nonlinear or even active load were applied, there would be distortions in the waveforms.

VI. CONCLUSION

Transitioning from high-intensity discharge (HID) lamps to LEDs in public lighting systems (PLS) offers substantial benefits in terms of energy efficiency and operational flexibility. However, ensuring accurate measurement and monitoring of energy consumption remains crucial to optimize these systems. Therefore, understanding the sources of measurement errors provides solutions to minimize their impacts on the measurements taken. Another important consideration is the choice of components used as sensors, as they will be the data collection interface, directly impacting the entire system.

The research conducted relied on MATLAB simulations to analyze the performance of the ADE9153A IC from Analog Devices under varying conditions. Thus, not utilizing the IC's autocalibration function leaves it susceptible to errors such as thermal drift, component variations, and gain inaccuracies. Conversely, leveraging the IC's autocalibration capabilities can mitigate errors related to offset, gain nonlinearity, and component variations, thereby improving the reliability of energy measurement in practical applications. It is also noteworthy that energy meters require factory calibration; therefore, auto-calibration helps reduce costs in the production line.

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TABLE II

COMPARISON OF REAL VALUES, CALIBRATED VALUES, AND ERRORS.

Parameter	Real Values	Measured Values	Error (%)
Voltage (Vrms)	220	219.999731	0.203760
Current (Irms)	5	5.010276	0.001878
Active Power (W)	1100.04	1115.923945	0.201330